

Score Normalization and Vote Construction Govern Preference-Graph Repair Outcomes in Multi-Ranker Retrieval

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Preference-graph repair is often motivated by the idea that removing cyclic inconsistency should improve downstream ranking quality, but that conclusion depends on how the graph is constructed. We study this question on SciDocs, FiQA, HotpotQA, and BRIGHT using stored BM25, TF-IDF, and MiniLM candidate scores, fixed candidate pools, and public qrels. Our primary analysis uses a normalized retention-matched protocol: per-query, per-ranker min-max normalization followed by retention-matched vote and edge thresholds. As a raw-score ablation, we also evaluate the unnormalized construction that treats native heterogeneous score margins as directly comparable. That unnormalized construction is severely scale dominated: when BM25 participated in an edge, its conditional mean weight share was 0.988, versus 0.512 under the primary protocol. Normalization and vote construction materially changed retained votes, edge weights, cyclicity, removed edges, and several repaired-versus-unrepaired signs. Under the primary protocol, repair still changes graph structure, but structural consistency and retrieval effectiveness remain distinct: repair can reduce graph inconsistency without producing a reliable improvement in ranked retrieval. No repaired-versus-unrepaired improvement remains robust after permutation testing, bootstrap uncertainty, influence analysis, and multiplicity correction. Simple baselines such as reciprocal rank fusion and CombSUM also remain strong comparators under the normalized graphs. The conclusion is therefore methodological as well as empirical: score normalization and vote construction jointly determine the preference graph on which repair operates, and claims about repair effectiveness must report those choices explicitly rather than treating graph construction as a neutral preprocessing step.

CCS Concepts: • **Information systems** → **Data cleaning**; **Data quality**; **Retrieval models and ranking**.

Additional Key Words and Phrases: preference graphs, data quality, structural inconsistency, rank aggregation, feedback arc set repair, retrieval evaluation, benchmark curation

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1 INTRODUCTION

Modern retrieval pipelines rarely rely on a single ranking signal. Sparse retrievers, dense retrievers, and other learned ranking components are often treated as separate sources of evidence whose outputs must be reconciled rather than trusted individually [1, 3, 6, 14]. Preference graphs offer a natural way to preserve that disagreement: for a given query, a directed edge from document u to document v records evidence that u should outrank v , and the edge weight determines how strongly that preference should influence the final ordering. This representation is appealing because it makes conflict explicit instead of collapsing it immediately into a single scalar score. But the graph is only as meaningful as the procedure used to construct it. If heterogeneous ranker scores are converted into edge weights without a defensible notion of comparability, then the resulting graph can reflect score scale as much as evidential quality.

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That issue matters because preference-graph repair operates entirely on the constructed graph. Once weighted pairwise evidence is aggregated, the graph may contain directed cycles, so no single ranking can satisfy every recorded preference. A standard response is to repair the graph by removing a minimum-weight feedback arc set before extracting a final ranking [1, 12]. This intervention improves structural consistency relative to the graph’s own edge weights, but structural consistency and retrieval effectiveness are not the same objective. A repair step can remove cycle-inducing edges, reduce a graph-internal inconsistency measure, and still fail to improve a relevance metric such as normalized discounted cumulative gain (nDCG) if the modified edges do not alter the final ranking in a helpful way. For that reason, a study of preference-graph repair must examine not only the repair algorithm but also the graph-construction choices that determine what the repair sees.

This paper studies that upstream dependency directly. Rather than asking for a predictive answer to “when repair helps,” we examine how score normalization and vote construction govern the preference graph on which repair operates, and how those choices affect both structural outcomes and downstream retrieval behavior. We evaluate four benchmarks—SciDocs, FiQA, HotpotQA, and BRIGHT—with stored BM25, TF-IDF, and MiniLM score files [4, 15, 19, 23]. Our primary protocol applies per-query, per-ranker min-max normalization followed by retention-matched thresholding; we retain the unnormalized raw-margin construction only as an ablation. The motivation for this comparison is empirical, not cosmetic. Under the raw-margin construction, BM25 contributed nearly all aggregate edge weight whenever it participated, so the graph was largely BM25-weighted even when three rankers were nominally present. Treating such raw heterogeneous margins as directly comparable support signals is not a neutral preprocessing choice; it is a substantive graph-construction assumption that shapes every downstream result.

Once graph construction is treated as a first-class variable, several measured outcomes change. Normalization and vote construction materially alter retained votes, edge weights, cyclicity, removed edges, and some repaired-versus-unrepaired signs. They also sharpen the distinction between total cyclicity and the smaller amount of nontrivial cyclicity that remains after direct mutual pairs are deleted. Repair therefore remains structurally active under the normalized graphs, but its retrieval effects are method-dependent and construction-sensitive rather than stable properties of “repair” in the abstract. The central empirical result is not that one dataset provides a robust positive exception, nor that repair has a construction-independent null effect. It is that graph-construction choices substantially determine both the structure being repaired and the retrieval conclusions drawn from that repair.

This framing changes how the paper’s contributions should be understood. We do not present feedback-arc-set repair as a generally superior reranking strategy, and we do not interpret raw-margin results as primary evidence. Instead, the paper contributes a normalized evaluation protocol for multi-ranker preference graphs, a structural analysis that separates mutual pairs from longer-cycle behavior, a four-dataset assessment of repaired versus unrepaired retrieval effects under paired statistical testing and robustness checks, and a raw-versus-normalized ablation showing that some repaired- versus-unrepaired signs are not stable under defensible graph reconstruction. These contributions support a more limited but more defensible conclusion: preference-graph repair outcomes are not invariant to score normalization and vote construction, and structural consistency should not be assumed to translate into retrieval improvement without direct evaluation.

The remainder of the paper is organized as follows. Section 2 positions the study relative to data-quality frameworks, rank aggregation, and rank fusion. Section 3 formalizes preference graphs, calibrated vote construction, structural metrics, and feedback-arc-set repair. Section 4 describes the four-dataset protocol, vote-extraction regimes, baselines, and statistical evaluation policy. The results sections then report calibrated structural behavior, repaired-versus-unrepaired retrieval effects, raw-versus-calibrated ablations, and the implications of those findings for baseline comparison and

practical use. The final sections discuss limitations, reproducibility, and the broader lesson that graph-construction choices must be reported explicitly in any empirical claim about preference-graph repair.

2 BACKGROUND AND RELATED WORK

Preference-graph methods sit at the intersection of several literatures: rank aggregation, noisy pairwise-comparison ranking, retrieval fusion, and data quality for derived analytical artifacts. The goal of this paper is narrower than a new ranking-method contribution. We do not introduce a new feedback-arc-set solver, a new retrieval ranker, or a new fusion rule. Instead, we study how heterogeneous ranker scores are converted into weighted preference graphs and how those graph-construction choices govern the measured effect of repair on both structural consistency and downstream retrieval.

Rank aggregation and preference graphs. Preference graphs are a standard representation for aggregating pairwise evidence across voters, rankers, or judges [1, 6, 17]. For a query-specific candidate set, a directed edge $u \rightarrow v$ records support for ranking document u ahead of document v , while the edge weight records the strength of that support. This representation is useful because it preserves disagreement as explicit graph structure instead of collapsing it immediately into a single fused score.

Feedback arc set repair and Kemeny-style consistency. Once a preference graph contains a directed cycle, no total ranking can satisfy every recorded preference simultaneously. Kemeny’s classical consensus ranking minimizes total pairwise disagreement with input preferences [11]; deciding an optimal Kemeny ranking and computing a minimum feedback arc set are closely related combinatorial problems [1, 6]. Weighted feedback arc set is NP-hard [10], so algorithms either optimize exact formulations on moderate instances or use heuristics. Classical work therefore studies minimum weighted feedback arc set (MWFAS) and Kemeny-style objectives that remove or violate as little edge weight as possible to obtain an acyclic ordering [1, 6, 8, 12]. That optimization target is purely graph-internal. It formalizes *structural* consistency, not relevance to external retrieval judgments. This distinction is central here: a better solution to the graph’s own repair objective need not improve a metric such as nDCG.

Probabilistic ranking from noisy pairwise preferences. Classical Bradley–Terry models treat pairwise outcomes as noisy comparisons about latent item strengths [2]. Rank Centrality likewise models pairwise outcomes as noisy evidence about a latent ordering and derives node scores from the induced comparison graph [17]. Those probabilistic methods are useful here as graph-dependent baselines; graph-repair methods instead remove edges to enforce acyclicity rather than fitting a probabilistic ranking model. Both families share the same upstream dependence: the object they consume is the constructed preference graph, so changing vote extraction or edge weighting can alter their behavior even when the upstream retrieval runs are unchanged.

Rank fusion in information retrieval. Information-retrieval systems already combine heterogeneous signals through score or rank fusion, including reciprocal rank fusion (RRF), CombSUM, and Borda-style methods [5, 9, 13]. Analyses of multiple-evidence combination clarify how score- and rank-based fusion behave when several retrieval strategies contribute overlapping evidence [13]. These baselines matter here for two reasons. First, they provide graph-free comparators against which repaired and unrepaired graph methods must be judged. Second, they make explicit a design choice that preference-graph work can hide: when heterogeneous retrieval scores are merged, some form of scale handling or an explicit decision to ignore cross-model score comparability becomes part of the method before score magnitudes are interpreted jointly. We keep CombSUM as the score-summation baseline and do not expand the primary baseline family with CombMNZ (an exploratory stored-score CombMNZ check did not overturn CombSUM’s

role enough to justify expanding the baseline table); the study’s comparative focus is repair sensitivity under fixed pools rather than an exhaustive fusion bake-off.

Score normalization for heterogeneous retrieval signals. BM25, TF-IDF, and neural similarity models operate on different native score scales. A raw difference of 0.05 has no reason to represent comparable evidential strength across those models, a point long recognized in meta-search score-normalization work [16]. The methodological issue studied in this paper is therefore not whether preference graphs are useful in principle, but whether weighted graph construction is stable when native heterogeneous score margins are treated as if they were directly comparable. Our primary normalized protocol makes that assumption explicit and testable rather than leaving it implicit in the edge weights.

LLM pairwise ranking and inconsistency. Recent work has used large language models as retrieval judges, rerankers, and pairwise preference sources [18, 20]. That line of work makes inconsistency salient, but it does not remove the measurement problem studied here. Whether the pairwise evidence comes from classical rankers or from LLM judgments, downstream claims about “repair” still depend on how pairwise evidence is filtered, weighted, and aggregated into a graph.

Derived-artifact data quality. We treat the preference graph itself as a derived data artifact whose properties can be audited. Cyclicity, strongly connected components, mutual contradictions, and removed feedback-arc weight are intrinsic properties of the graph produced by a construction pipeline, whereas retrieval metrics such as nDCG are extrinsic properties of rankings evaluated against qrels. The contribution of this paper is to separate those two measurement families and show that conclusions about repair are not invariant to score normalization and vote construction. In this sense, the paper is a study of graph construction and measurement sensitivity, not a claim that a particular repair heuristic solves retrieval reranking.

3 METHODOLOGY

This section defines the calibrated preference-graph pipeline used for the primary analysis. The order matters. Upstream score scale affects pairwise margins; pairwise margins affect retained votes; retained votes determine weighted graph edges; graph structure determines what repair sees; and repair influences ranking extraction only through the modified graph. Table 1 summarizes notation used in this section and Section 4, and Figure 1 summarizes the end-to-end pipeline these subsections define, stage by stage.

3.1 Upstream Scores and Candidate Pool

The primary pipeline starts from stored score files for three upstream rankers: BM25, TF-IDF, and MiniLM. For each query q , we first define a shared candidate pool D_q from the union of documents scored by at least one ranker. If that union already has size at most the dataset-specific pool size, we keep all documents. Otherwise we rank the union by reciprocal rank fusion with parameter $k = 60$ over the full per-ranker native ranked lists and keep the top- $|D_q|$ candidates. Candidate pooling therefore uses an RRF-family computation, but it is distinct from both the Prior ranking and the reported RRF baseline because it is applied before graph construction and before candidate restriction.

3.2 Per-Query, Per-Ranker Score Calibration

Throughout this paper, “calibration” and “calibrated” denote the specific per-query, per-ranker min–max normalization and retention-matching protocol defined in this subsection and Section 3.3, used as manuscript-specific shorthand. We do not use the term in the probabilistic sense (e.g., Platt scaling or isotonic regression, where a calibrated score is

Table 1. Notation used in Sections 3 and 4.

Symbol	Meaning
q, D_q	Query and its candidate document set
s	Upstream ranker in {BM25, TF-IDF, MiniLM}
$x_{q,s}(d)$	Native score assigned by ranker s to document d for query q
$\bar{x}_{q,s}(d)$	Calibrated per-query, per-ranker score
$m_{q,s}(u, v)$	Absolute pairwise margin for pair $\{u, v\}$ under ranker s
$G_q = (V_q, E_q, w_q)$	Preference graph for query q ($V_q = D_q$)
$\hat{G}_q$	Repaired (acyclic) preference graph
r	Vote-extraction regime, $r \in \{\text{ms2}, \text{ms1}, \text{ms1_drop_mutual}\}$
κ_r	Regime-specific minimum support threshold
λ_s	Per-ranker vote-retention threshold within one dataset/protocol/regime cell
γ_r	Aggregate edge-retention threshold within one dataset/protocol/regime cell
$c_q(u, v)$	Support count for direction $u \rightarrow v$
$a_q(u, v)$	Aggregate retained weight for direction $u \rightarrow v$
F_{opt}	Exact minimum-weight feedback arc set (optimization target)
F_{greedy}	Greedy removed-edge set computed in the experiments
$\deg^+(u), \deg^-(u)$	Out-degree, in-degree of node u in a graph
$s_{\text{cop}}, s_{\text{bal}}, s_{\text{mc}}$	Copeland, weighted balance, and Markov graph scores
$\hat{s}(\cdot)$	Min-max normalization of a score vector to $[0, 1]$
α	Hybrid mixing weight for the graph-derived component
π	A ranking (permutation) of V_q
$\text{BEW}(\pi, G)$	Backward-edge weight of π relative to graph G
$\text{PIC}(\pi, G)$	Pairwise inconsistency count (unweighted analogue)
k	nDCG cutoff

claimed to match an empirical outcome frequency), and no such claim is made or implied anywhere in this paper: the transformed scores are ordinal, per-query, per-ranker rescalings to $[0, 1]$, not probability estimates.

The primary protocol does not treat native BM25, TF-IDF, and MiniLM margins as directly comparable edge-weight contributions. For each query q and ranker s , let $D_{q,s} \subseteq D_q$ be the candidate documents for which ranker s supplies a score. We define the calibrated score

$$\bar{x}_{q,s}(d) = \begin{cases} 0, & \max_{d' \in D_{q,s}} x_{q,s}(d') - \min_{d' \in D_{q,s}} x_{q,s}(d') \leq 10^{-12}, \\ \frac{x_{q,s}(d) - \min_{d' \in D_{q,s}} x_{q,s}(d')}{\max_{d' \in D_{q,s}} x_{q,s}(d') - \min_{d' \in D_{q,s}} x_{q,s}(d')}, & \text{otherwise.} \end{cases} \quad (1)$$

This transformation is applied separately for each query and ranker after candidate restriction. Missing documents remain missing: if a ranker did not score document d , no calibrated score is imputed for that document. If a ranker's candidate-restricted score vector is constant, every calibrated score is set to 0 and that ranker contributes no nonzero within-query margin.

Calibration is necessary because the edge weights in a multi-ranker preference graph inherit whatever comparability assumptions are imposed on the upstream scores. We therefore treat the unnormalized raw-margin construction only as an ablation and use Eq. (1) as the primary graph-construction input.

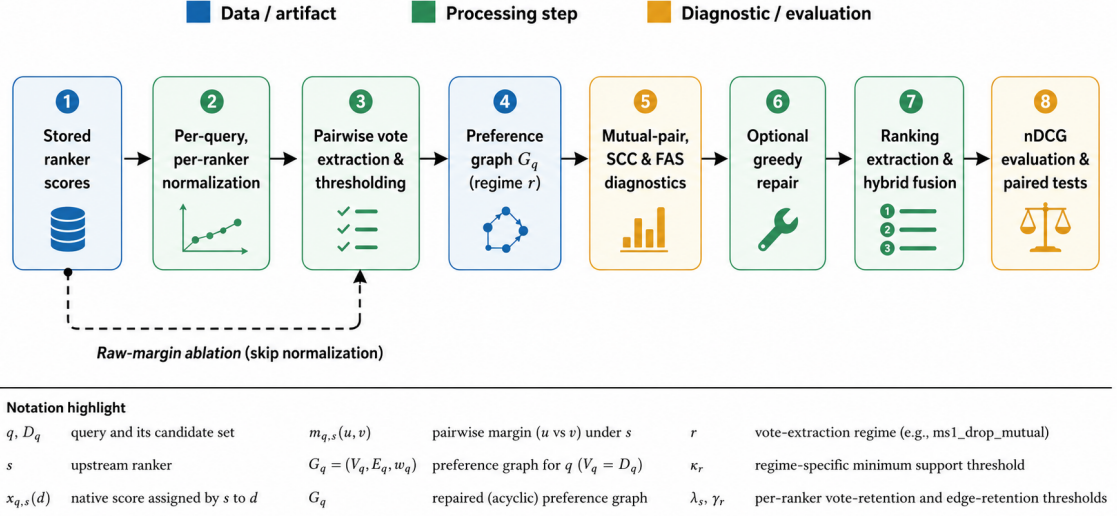


Fig. 1. The eight-stage evaluation pipeline (numbered 1–8), color-coded by role: blue stages are data artifacts, green stages are processing steps, and amber stages are diagnostic or evaluation steps. Stored ranker scores are normalized per query and per ranker (Eq. (1)), converted to pairwise votes and thresholded into a preference graph G_q under regime r , diagnosed structurally, optionally repaired, and finally converted to a ranking and scored by nDCG under paired statistical tests. The dashed arrow marks the raw-margin ablation, which reuses every downstream stage unchanged but skips normalization and keeps fixed raw-margin thresholds. A notation-highlight key below the diagram repeats the symbols most relevant to this pipeline from Table 1. The reader should take away that normalization and repair are two distinct, separated stages, not one combined step.

3.3 Pairwise Vote Extraction and Graph Construction

For each query q , ranker s , and unordered candidate pair $\{u, v\} \subseteq D_q$, the primary protocol compares calibrated scores only if both documents are present in $D_{q,s}$. If either document is missing, ranker s abstains on that pair. If $\bar{x}_{q,s}(u) = \bar{x}_{q,s}(v)$, ranker s also abstains. Otherwise the absolute pairwise margin is

$$m_{q,s}(u, v) = |\bar{x}_{q,s}(u) - \bar{x}_{q,s}(v)|. \quad (2)$$

Ranker s casts a directional vote only when $m_{q,s}(u, v) \geq \lambda_s$, where λ_s is the per-ranker vote-retention threshold for the current dataset, protocol, and regime. The direction is given by the larger calibrated score. For a candidate direction $u \rightarrow v$, let

$$c_q(u, v) = \sum_s 1[s \text{ retains } u \rightarrow v] \quad (3)$$

be the support count and

$$a_q(u, v) = \sum_s m_{q,s}(u, v) 1[s \text{ retains } u \rightarrow v] \quad (4)$$

be the aggregate directional weight. A direction is retained only if

$$\begin{aligned} c_q(u, v) &\geq \kappa_r, \\ a_q(u, v) &\geq \gamma_r, \end{aligned} \quad (5)$$

where κ_r is the regime-specific minimum support and γ_r is the aggregate threshold for the current dataset/protocol/regime cell. The graph builder then sums all retained per-ranker contributions for the same directed pair, so the final edge weight is $w_q(u, v) = a_q(u, v)$ whenever Eq. (5) is satisfied. Because each direction is tested independently, a single unordered pair can retain both $u \rightarrow v$ and $v \rightarrow u$.

3.4 Vote-Extraction Regimes

We evaluate three vote-extraction regimes over the same stored score files and candidate pools:

- **ms2** requires support $\kappa_r = 2$ and applies a nontrivial aggregate threshold γ_r . Under the unnormalized raw ablation this threshold is 0.1; under the primary protocol it is retention-matched separately for each dataset.
- **ms1** requires support $\kappa_r = 1$ and uses the dataset-specific aggregate threshold for that protocol/regime cell. It preserves directional disagreement whenever both directions independently survive thresholding.
- **ms1_drop_mutual** uses the same support rule as ms1, but after thresholding deletes every unordered pair that retained both directions. It therefore treats direct bidirectional contradiction as unresolved rather than retaining either arc.

The interpretation of these regimes is important. Cyclicity under ms1 does not automatically imply longer-cycle nontransitivity; it can arise from direct mutual contradictions alone. We therefore separate mutual pair diagnostics from residual cyclic structure after mutual-pair deletion.

3.5 Retention-Matched Thresholds

Under the primary protocol, thresholds are not fixed globally at the raw values 0.05 and 0.1. Instead, for each dataset, regime, and calibration variant, we match retention behavior to the unnormalized raw-margin construction without using qrels. Let $\rho_{s,r}^{\text{raw}}$ denote the raw retained vote rate for ranker s in regime r , computed over the stored query subset with denominator equal to the number of candidate pairs for which both documents are scored by s and the native scores are unequal. For a calibration variant, we pool all calibrated margins $m_{q,s}(u, v)$ across the dataset and set

$$\lambda_s = Q_{1-\rho_{s,r}^{\text{raw}}}(\{m_{q,s}(u, v)\}), \quad (6)$$

where Q_p denotes the empirical quantile at level p . If a ranker has no eligible margins in that cell, we set $\lambda_s = 0$.

After vote-threshold matching, we choose the aggregate threshold γ_r deterministically to minimize the absolute gap between the retained edge count under the calibrated construction and the retained edge count under the raw reference construction for the same dataset and regime. The search is over the realized aggregate candidate weights in that cell; when multiple thresholds yield the same minimum gap, the smaller threshold is selected. Qrels are not used at any stage of this threshold selection procedure. This matching strategy keeps graph density and vote-retention behavior approximately comparable across calibration variants without tuning for nDCG.

Retention matching is an experimental control, not a claim about optimal threshold selection. Its sole purpose is to let a raw-versus-calibrated comparison isolate the effect of score-scale handling: if thresholds were left unmatched, any structural or retrieval difference between the raw and calibrated protocols could be confounded by unrelated differences in graph density or edge count rather than by score-scale handling itself. This procedure is therefore neither a universally optimal threshold-selection method nor a qrels-tuned one, and we do not present it as either. A sensitivity sweep of nearby retention/threshold policies (Section 9) shows that graph density, cyclicity, and removed-edge patterns can change, and that some repaired-versus-unrepaired cell signs flip, while no positive cell survives Holm correction

under those alternatives. We therefore retain raw-reference retention matching as the primary protocol rather than treating any single nearby target as universally optimal.

3.6 Mutual Pairs and Nontrivial Cycles

We distinguish two kinds of cyclic structure. A *mutual pair* is a direct contradiction in which both $u \rightarrow v$ and $v \rightarrow u$ are retained. A *nontrivial cycle* is any directed cycle that remains after deleting all mutual-pair edges. The primary results therefore report cyclic query percentage and largest strongly connected component (SCC) size both before and after mutual-pair deletion.

3.7 Structural Diagnostics

We group structural measurements into intrinsic graph diagnostics and qrels-anchored diagnostics.

Intrinsic graph diagnostics. For each query graph we record edge count, graph density $|E_q|/(|D_q|(|D_q| - 1))$, mutual-pair count, the share of queries containing any mutual pair, cyclic-query indicators before and after mutual-pair deletion, largest SCC size before and after mutual-pair deletion, directed triangle counts, total graph weight $W_q = \sum_{(u,v) \in E_q} w_q(u, v)$, and normalized graph weight $W_q/\max(1, |E_q|)$. After repair we also record the number of removed edges, their total removed weight, and normalized removed weight W_q^{rem}/W_q when $W_q > 0$ and 0 otherwise.

Qrels-anchored diagnostics. We additionally report two diagnostics that compare a graph to an external reference ranking π_{ref} derived from relevance judgments: the **backward-edge weight**

$$\text{BEW}(\pi_{\text{ref}}, G) = \sum_{(u,v) \in E: \pi_{\text{ref}}(v) < \pi_{\text{ref}}(u)} w(u, v), \quad (7)$$

and the **pairwise inconsistency count**

$$\text{PIC}(\pi_{\text{ref}}, G) = |\{(u, v) \in E : \pi_{\text{ref}}(v) < \pi_{\text{ref}}(u)\}|, \quad (8)$$

the unweighted count of the same contradictions. These are not intrinsic graph properties and are not independent validation signals: they use the same qrels family as downstream retrieval evaluation. We therefore interpret them as reference-alignment diagnostics rather than as separate evidence of retrieval quality.

3.8 Graph Repair

The weighted feedback arc set optimization problem for graph G_q is

$$F_{\text{opt}} = \arg \min_{F \subseteq E_q : (V_q, E_q \setminus F) \text{ acyclic}} \sum_{(u,v) \in F} w_q(u, v), \quad (9)$$

with repaired graph $\tilde{G}_q = (V_q, E_q \setminus F, w_q)$ when a set F is removed. The optimizer need not be unique; when multiple minimum-weight removal sets exist, any one optimal set is valid for the objective. In the primary experiments we do not solve Eq. (9) exactly. Instead, we compute a greedy removal set F_{greedy} by repeatedly finding a directed cycle in the current graph, removing the minimum-weight edge from that discovered cycle, and continuing until no cycle remains. The implementation uses `networkx.find_cycle` to identify a cycle at each step. This greedy cycle-peeling procedure is related in spirit to classical feedback-arc-set heuristics that remove edges based on local cycle structure [7], but it is a simple cycle-peeling variant rather than a reimplement of that specific algorithm, and we claim no worst-case approximation bound for it.

This procedure approximately minimizes removed edge weight through greedy cycle peeling, but it has no approximation guarantee and its output can depend on cycle-discovery order. We therefore report removed weight and removed-edge overlap empirically, without labeling the computed solution as an exact minimum-weight feedback arc set. Repair remains a graph-internal operation: it depends only on w_q , not on qrels or nDCG.

For the exact robustness check in Section 4.4, we solve Eq. (9) with a linear-ordering MIP. For distinct nodes $u, v \in V_q$, let $x_{uv} \in \{0, 1\}$ mean that u precedes v in the recovered order. The program uses

$$x_{uv} + x_{vu} = 1 \quad \text{for all unordered pairs } \{u, v\}, \quad (10)$$

$$x_{ab} + x_{bc} + x_{ca} \leq 2 \quad \text{for all distinct triples } a, b, c, \quad (11)$$

$$\min \sum_{(u,v) \in E_q} w_q(u, v) x_{vu}. \quad (12)$$

An edge (u, v) is treated as reversed—and therefore removed—when $x_{vu} = 1$. The reported exact comparison uses SCIP via PySCIPOpt 6.2.1 with MIP gap 0, accepts only proven-optimal status, and applies a 300 s per-query safety limit that was never reached on the 1,025 primary-protocol graphs (all solved to proven optimality; mean solve time 7.4 ms). The 87.9% disagreement rate and 26.3% lower removed weight are computed on the 379 cyclic queries only, comparing removed-edge sets and total removed weight between exact and greedy repairs. No graph was excluded for timeout or non-optimal status.

3.9 Ranking Extraction

Ranking extraction is applied separately to G_q and $\tilde{G}_q$. Graph construction and hybrid normalization must be kept distinct: the calibration in Eq. (1) is applied before graph construction, whereas the normalization in Eq. (15) is applied only when combining graph-derived scores with a prior ranking.

We evaluate three graph-derived ranking families. The **Copeland score** is out-degree minus in-degree,

$$s_{\text{cop}}(u) = \deg^+(u) - \deg^-(u), \quad (13)$$

counting net wins irrespective of edge weight. The **weighted balance score** is the signed sum of incident edge weight,

$$s_{\text{bal}}(u) = \sum_{(u,v) \in E} w(u, v) - \sum_{(v,u) \in E} w(v, u). \quad (14)$$

The **Markov score** s_{mc} is the stationary distribution of a Rank-Centrality-style Markov chain on the weighted preference graph [17]. We use teleportation probability 0.15 (equivalently, retention probability 0.85), which yields an irreducible and aperiodic chain and therefore a unique stationary distribution even when the underlying preference graph is cyclic or disconnected. Final rankings sort by descending score with deterministic tie-breaking by document id when scores are equal; the Markov ranking additionally breaks ties by smaller weighted in-degree before document id.

The Prior ranking is a candidate-restricted RRF score over the stored ranker lists with parameter $k = 60$. The reported RRF baseline also uses $k = 60$ on the candidate-restricted lists, but its tie-breaking is not identical to Prior. We therefore name those roles separately rather than treating Prior and RRF as the same object.

The primary hybrid methods combine Prior with either Copeland or balance. Writing $\hat{s}(\cdot)$ for min-max normalization of a score vector to $[0, 1]$ over the candidate set, the hybrid score is

$$s_{\text{hybrid}}(u) = \hat{s}_{\text{prior}}(u) + \alpha \hat{s}_{\text{comp}}(u), \quad \text{comp} \in \{\text{cop}, \text{bal}\}, \quad (15)$$

with $\alpha = 0.3$ in the main comparisons. The primary analysis additionally reports a small alpha sensitivity sweep over $\alpha \in \{0.1, 0.3, 0.5, 1.0\}$ for this primary protocol only.

3.10 Retrieval Evaluation

Retrieval quality is measured by nDCG at the dataset-specific cutoff used in candidate pooling. For a ranking π and a relevance map $\text{rel} : D_q \rightarrow \mathbb{Z}_{\geq 0}$,

$$\text{nDCG}@k(\pi) = \frac{\sum_{i=1}^k \frac{2^{\text{rel}(\pi_i)} - 1}{\log_2(i+1)}}{\sum_{i=1}^k \frac{2^{\text{rel}(\pi_i^{\text{ideal}})} - 1}{\log_2(i+1)}}, \quad (16)$$

where π_i is the document at rank i in π and π^{ideal} sorts candidate documents by descending relevance. $\text{nDCG}@k$ is defined as 0 when the denominator is 0. All paired comparisons use only the intersection of queries for which both methods produce valid outputs.

3.11 Raw-Margin Ablation

The unnormalized raw-margin protocol is retained only as an explicitly labeled ablation. It uses native BM25, TF-IDF, and MiniLM scores without cross-ranker normalization; a fixed per-vote threshold of 0.05 for all three rankers; and fixed aggregate/support settings: ms2 uses minimum support 2 and aggregate threshold 0.1, while ms1 and ms1_drop_mutual use minimum support 1 and aggregate threshold 0.0. We do not interpret those raw margins as comparable confidence weights.

4 EXPERIMENTAL SETUP

4.1 Datasets and Stored Query Sets

We use four public retrieval benchmarks spanning distinct domains: SciDocs (scientific document retrieval) [4], FiQA (financial opinion and question answering) [15], HotpotQA (multi-hop question answering) [23], and BRIGHT (reasoning-intensive retrieval) [19]. SciDocs, FiQA, and HotpotQA are also part of BEIR [21]. The calibrated rerun does not draw a fresh query sample. It reuses the fixed stored query_ids.txt files from the primary experiment inputs and then filters them only for evaluation eligibility.

A query is eligible if its qrels support ranking comparison: either at least two judged documents with at least two distinct relevance grades, or at least one judged document with strictly positive relevance. Queries failing this criterion are excluded before normalization, graph construction, and evaluation. The filter uses only qrels; it does not depend on method rankings or repair outcomes, and the same eligible query set is used for every compared protocol and method. On SciDocs, FiQA, and BRIGHT every stored query is eligible. On HotpotQA, 18 of the 70 stored IDs have only nonpositive relevance labels under the processed qrels and are therefore excluded, leaving 52 usable queries. Table 2 reports the stored query subset size, the usable query count under the primary protocol, the upstream retrieval depth used for stored score generation, the candidate-pool size, and the nDCG cutoff.

4.2 Upstream Rankers and Candidate Pooling

The upstream score files are fixed across all protocols and regimes. They come from BM25, TF-IDF, and MiniLM and are never regenerated in the calibrated study. Candidate pooling uses the union of documents appearing in those score

Table 2. Stored query subsets and evaluation settings reused by the four-dataset primary analysis.

Dataset	Stored IDs	Usable queries	Top- n	$ D_q $	nDCG@ k
SciDocs	120	120	50	20	20
FiQA	120	120	50	20	20
HotpotQA	70	52	35	10	10
BRIGHT	50	50	50	20	20

files for a query; if the union exceeds the dataset-specific pool size, it is truncated by RRF with $k = 60$ over the full native ranked lists, following the conventional RRF constant used by Cormack et al. and common retrieval toolkits such as Anserini/BEIR fusion utilities [5]. The study therefore evaluates a specific *RRF-centered candidate-pool setting*. It does not claim candidate-pool neutrality across alternative pool construction policies.

The study uses exactly the same stored score files, candidate pools, and qrels for all protocols. Differences among ms2, ms1, and ms1_drop_mutual therefore reflect only vote extraction and graph construction, not changes in upstream evidence.

4.3 Methods and Baselines

The calibrated package evaluates four graph-independent baselines and ten graph-dependent methods. The graph-independent baselines are Prior, RRF, CombSUM, and Borda fusion. The graph-dependent methods are unrepaired and repaired variants of Copeland, balance, and Markov extraction, plus unrepaired and repaired Copeland and balance hybrids at $\alpha = 0.3$.

CombSUM uses per-query, per-ranker min-max normalization before summation, so missing documents contribute 0 for that ranker. Borda fusion operates on the rank positions of the stored score lists over the union of scored documents for that query. Prior and the RRF baseline both use the RRF family with $k = 60$ on candidate-restricted lists, but they are not bitwise-identical implementations: Prior ranks by fused prior score with document-id tie-breaking, whereas the RRF baseline breaks ties by best observed rank and then document id. That tie-breaking difference alone yields identical full rankings in only 216 of 6,156 comparable Prior-RRF cases in Section 6, so we report both and treat them as related but non-interchangeable baselines. These classical fusion baselines serve as graph-independent controls for the repair-sensitivity question; the study does not aim to establish absolute retrieval leadership against modern cross-encoder or LLM rerankers, which would require a different experimental design with new inference runs and are outside the measurement contribution pursued here.

The graph-independent baselines are identical across vote regimes for a fixed query because they operate directly on the stored score files rather than on the constructed preference graph. Regime-labeled baseline rows are reported only to align comparisons with graph-dependent methods; they are not independent reruns.

4.4 Repair Configuration

All primary results use the greedy cycle-peeling repair described in Section 3.8. To check whether this heuristic’s known sub-optimality changes either the structural or the retrieval conclusions, we additionally solved the exact minimum-weight feedback arc set problem (Eq. (9)) with an open-source mixed-integer-programming solver (SCIP via PySCIPOpt 6.2.1; no commercial solver dependency) on every query in the primary normalized protocol—all 1,025 queries across all four datasets and three vote regimes, not a bounded sample. Every solve was accepted only when

Table 3. Method families evaluated in the calibrated four-dataset package.

Method family	Uses graph?	Repaired and unrepaired?
Prior	No	—
Reciprocal rank fusion (RRF)	No	—
CombSUM	No	—
Borda-count fusion	No	—
Copeland graph	Yes	Yes
Balance graph	Yes	Yes
Markov graph	Yes	Yes
Copeland hybrid (Eq. (15))	Yes	Yes
Balance hybrid (Eq. (15))	Yes	Yes

Table 4. Repair configurations used in the primary analysis and robustness check.

Configuration	Role
Greedy cycle peeling	Primary repair for normalized and raw-ablation runs, all 1,025 queries
Exact MWFAS (SCIP / PySCIOpt 6.2.1)	Robustness check only, same 1,025 queries; confirms greedy’s structural sub-optimality but no change to the retrieval conclusion

SCIP reported proven-optimal status with MIP gap 0; solutions were independently cross-checked against brute-force exact enumeration on 49 synthetic and real graphs. Table 4 reports both configurations. Among the 379 queries whose raw graph actually contains a cycle, the exact solver selects a different removed-edge set than greedy in 87.9% of them and removes 26.3% less total edge weight on average, confirming that greedy cycle peeling is measurably, not just theoretically, sub-optimal on this data. This structural difference does not, however, change the retrieval conclusion: across every repaired-graph-dependent method and $n\text{DCG}@ \{5, 10, 20\}/\text{MRR}/\text{MAP}$, none of 35 pooled or 399 per-dataset-regime exact-minus-greedy cells survives Holm correction. We therefore report the exact solver only as a robustness check on the retrieval conclusion, not as a second primary repair configuration, and continue to use greedy cycle peeling for all primary $n\text{DCG}$ analyses.

4.5 Statistical Analysis

All repaired-versus-unrepaired comparisons are paired at the query level. For each cell we report the mean delta, median delta, 5th/25th/75th/95th-percentile query-level deltas, and helped/harmed/unchanged query counts. We also report a paired sign-flip permutation test and a paired percentile bootstrap interval with 10,000 resamples, treating significance, interval uncertainty, and multiplicity-corrected decisions as distinct layers of evidence [22]. If δ_i denotes the per-query repaired minus unrepaired $n\text{DCG}$ difference for a paired set of n queries, and $\bar{\delta}^{(b)}$ is the mean of bootstrap resample b , the reported percentile interval is

$$\left[\bar{\delta}^{(\lceil 0.025(B-1) \rceil)}, \bar{\delta}^{(\lceil 0.975(B-1) \rceil)} \right], \quad (17)$$

with $B = 10,000$. Bootstrap summaries additionally include the fraction of bootstrap means greater than zero.

The primary inferential family is the set of repaired-versus-unrepaired $n\text{DCG}$ cells under the min-max plus retention-matched protocol. We apply both Holm and Benjamini-Hochberg correction across that family. We do not treat a bootstrap interval endpoint equal to zero as evidence of a strictly positive effect, and we do not infer robustness

from percentile intervals alone. For interpretation-sensitive cells we additionally report leave-one-query-out and top-influence-removal analyses.

As a secondary robustness check, we also recompute MAP and mean reciprocal rank (MRR) from the same stored candidate scores, pools, and rankings used for nDCG, without regenerating upstream retrieval. Those binary-relevance metrics are exploratory relative to the primary nDCG family: they are not folded into the main Holm/BH correction, and pool-truncated Recall@ k is omitted because candidates are already restricted before ranking. Across MAP and MRR, no repaired-versus-unrepaired cell survives Holm correction; a small mixed-sign BH subset remains (especially SciDocs ms1 positive and BRIGHT ms1 negative), so the secondary metrics do not overturn the primary nDCG conclusion.

4.6 Implementation and Reproducibility

The primary analysis pipeline is implemented in Python 3.12.3 using networkx for graph operations. Run manifests record the executable, stored score-file hashes, qrels hashes, query-ID lists, protocol definitions, threshold tables, and generated outputs. Because this is an anonymous review draft and no verified anonymized artifact URL was available at writing time, we describe the artifact contents here rather than inserting a public author-identifying repository link.

The primary analysis is deterministic given the stored inputs and recorded seeds. In particular, it reuses fixed query-ID manifests instead of redrawing query samples. Bootstrap intervals use seed 13 with $B = 10,000$ resamples; paired sign-flip permutation tests use Monte Carlo sampling with 10,000 replications and seed 17 (not an exact 2^n enumeration). The study does not claim exact reproducibility for separate real-LLM judgment pipelines.

4.7 Real-LLM Evidence: Scope and Role

In addition to the mechanical three-ranker evaluation described above, we inspect stored real-LLM judgment corpora on subsets of SciDocs, HotpotQA, and FiQA, but we no longer treat those records as a parallel retrieval evaluation. Section 7.2 reframes them as a bounded protocol-quality check because the stored corpora differ from the mechanical study in prompt symmetry, parser behavior, exclusion logging, and provider coverage. We flag that distinction here because those corpora are easy to conflate with the main mechanical-vote evaluation if their weaker evidentiary status is not made explicit.

5 STRUCTURAL DATA QUALITY RESULTS

This section reports the primary normalized protocol introduced in Section 3.2: per-query, per-ranker min-max score normalization followed by retention-matched vote thresholds. The unnormalized raw-margin construction is retained only as an ablation for sensitivity analysis. We separate four questions that a raw-margin analysis alone conflates: score-scale imbalance, graph construction, the share of cyclicity attributable to direct mutual contradictions, and the amount of structural work repair actually performs.

5.1 Raw Score Scale Imbalance

The raw protocol is strongly dominated by native BM25 scale. Under the unnormalized raw-margin construction, BM25 contributes a mean conditional edge weight share of 0.988 whenever it participates in an edge; under the primary protocol, the corresponding mean falls to 0.512. This shift does not show that BM25 becomes less useful after normalization. It shows that the unnormalized weighted graph was largely tracking one ranker's score scale, not combining commensurate evidence from BM25, TF-IDF, and MiniLM. A supporting analysis of the fixed raw thresholds further shows

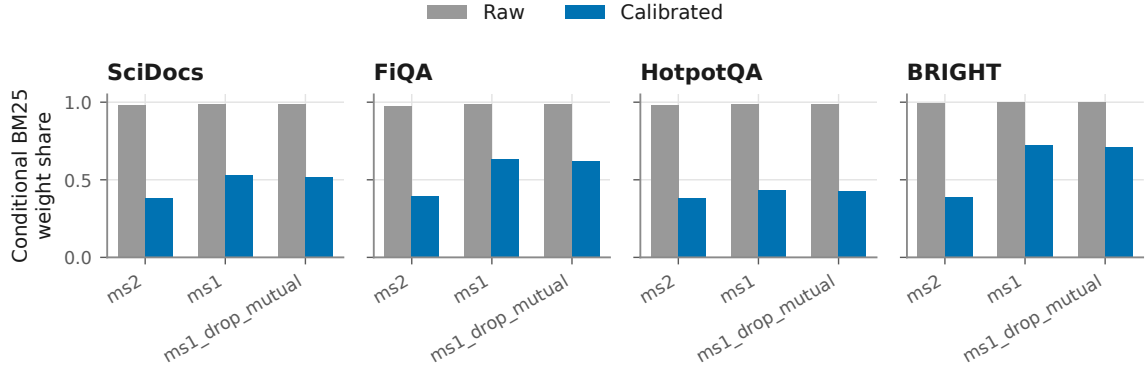


Fig. 2. Mean conditional BM25 edge-weight share by dataset, vote regime, and protocol: gray bars are the raw-margin ablation, blue bars are the primary normalized protocol (per-query/per-ranker min-max normalization with retention-matched thresholds). BM25 share is near 1.0 under the raw protocol in every panel and drops to roughly 0.4–0.7 after calibration, confirming that the raw graph was tracking BM25’s score scale rather than combining three rankers’ evidence. This is a graph-construction diagnostic, not a retrieval-quality measurement.

Table 5. Intrinsic graph diagnostics under the primary normalized protocol. Cyclic% is the share of query graphs containing at least one directed cycle before mutual-pair deletion; Post-mutual Cyclic% recomputes the same statistic after deleting direct mutual pairs; Largest SCC is the mean size of the largest strongly connected component before deletion; normalized FAS is mean removed repair weight divided by total graph weight. Raw-margin results are treated only as an ablation and are not shown as the main evidence here.

Dataset	Regime	Cyclic%	Post-mutual Cyclic%	Largest SCC	Norm. FAS
SciDocs	ms2	0.0	0.0	1.00	0.000
SciDocs	ms1	99.2	10.8	16.17	0.074
SciDocs	ms1_drop_mutual	11.7	11.7	1.93	0.002
FiQA	ms2	0.0	0.0	1.00	0.000
FiQA	ms1	98.3	30.8	16.12	0.080
FiQA	ms1_drop_mutual	30.8	30.8	3.38	0.004
HotpotQA	ms2	0.0	0.0	1.00	0.000
HotpotQA	ms1	63.5	1.9	4.06	0.029
HotpotQA	ms1_drop_mutual	1.9	1.9	1.13	0.000
BRIGHT	ms2	0.0	0.0	1.00	0.000
BRIGHT	ms1	92.0	22.0	10.82	0.055
BRIGHT	ms1_drop_mutual	22.0	22.0	3.08	0.003

they were not semantically comparable across rankers, so raw score margins cannot be interpreted as cross-ranker confidence weights. Figure 2 shows this shift directly.

5.2 Calibration Changes Graph Construction

Calibration changes more than the numeric scale of existing edges. It changes which edges are retained, how much total weight they carry, and which edges are ultimately selected for repair. The overlap between the primary protocol’s removed edges and the raw-margin ablation is only 0.090 on SciDocs ms1, 0.055 on FiQA ms1, 0.328 on HotpotQA ms1, and 0.110 on BRIGHT ms1. These low overlaps show that calibration changes repair decisions themselves, not merely the magnitudes assigned to an otherwise stable set of edges. Under the primary protocol, ms2 remains acyclic on all four datasets, while ms1 is still highly cyclic before mutual-pair deletion. Figure 3 summarizes that primary regime-level pattern and Figure 4 contrasts it with the raw-margin ablation.

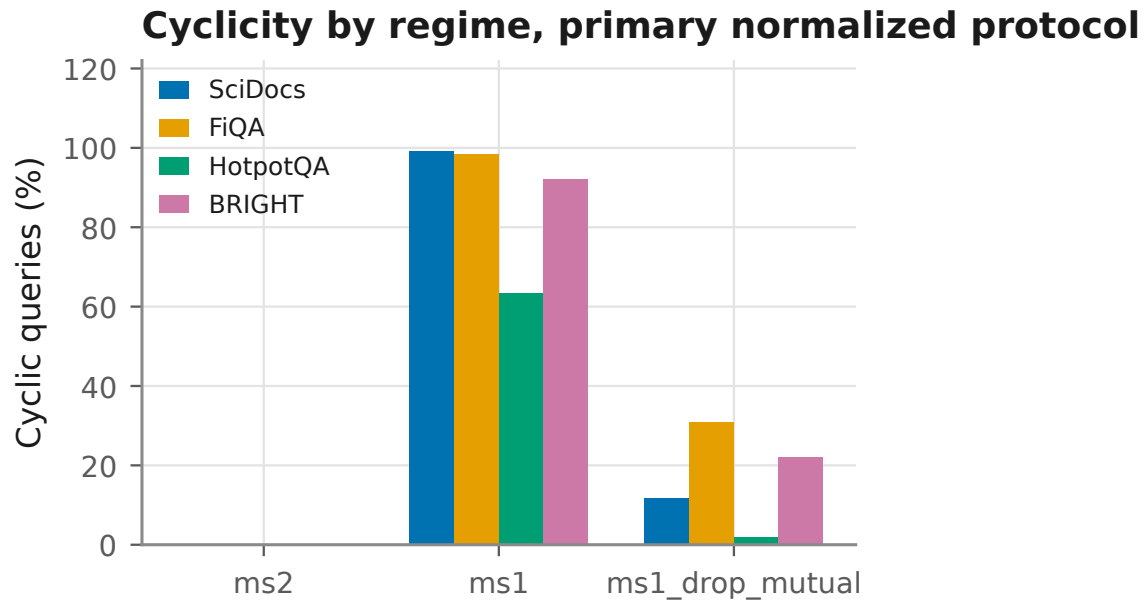


Fig. 3. Cyclic-query percentage under the primary normalized protocol, grouped by vote regime with one bar per dataset. ms2 is exactly 0% on every dataset; ms1 is 64–99%; deleting direct mutual pairs (ms1_drop_mutual) brings every dataset back down to 2–31%. The reader should notice that ms1’s high cyclicity is concentrated almost entirely in direct mutual contradictions, not longer cycles – Figure 5 isolates that residual directly. Values before mutual-pair deletion; post-deletion values are also in Table 5.

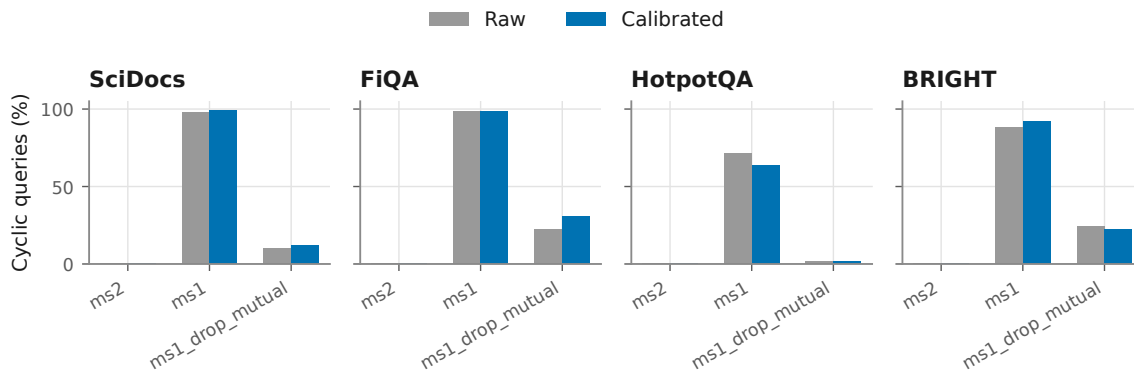


Fig. 4. Cyclic-query percentage by vote regime, raw-margin ablation (gray) versus primary normalized protocol (blue), one panel per dataset. Unlike the BM25 weight share in Figure 2, overall cyclic-query rate changes only modestly between raw and calibrated construction within each regime. The reader should not conclude that calibration leaves the graph unchanged: Section 5 shows the removed-edge overlap between raw and calibrated repair is only 0.055–0.328, so calibration changes *which* edges are cyclic and removed far more than it changes the headline cyclic-query rate shown here.

5.3 Mutual Pairs Versus Residual Nontrivial Cyclicity

The calibrated graphs remain cyclic under ms1, but a substantial share of that cyclicity comes from direct mutual contradictions rather than longer preference loops. Averaged over all dataset×regime cells in the primary protocol,

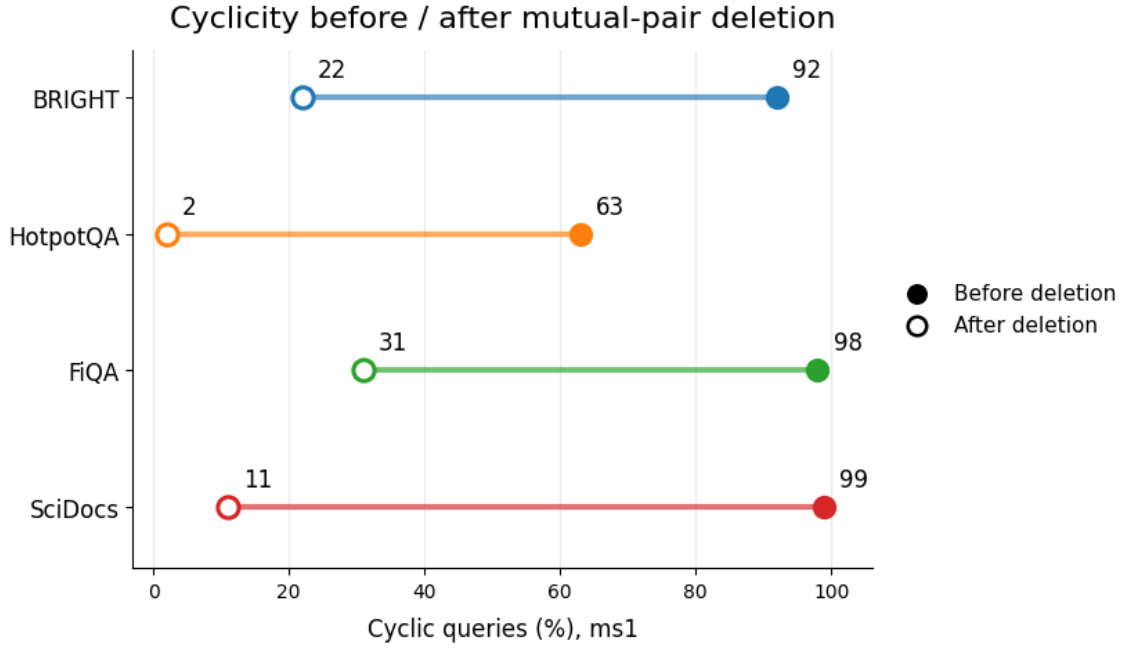


Fig. 5. Cyclic-query percentage under ms1 before (filled dot) and after (open dot) deleting direct mutual pairs, one row per dataset. The drop is largest for HotpotQA (63% $\rightarrow$ 2%) and smallest for FiQA (98% $\rightarrow$ 31%). The reader should take away that most ms1 cyclicity is direct bidirectional contradiction, not longer-cycle nontransitivity, and that the remaining share after deletion varies substantially by dataset.

cyclic-query prevalence is 0.3495 before mutual-pair deletion and 0.1100 afterward. The contrast is especially sharp on HotpotQA: its ms1 cyclic-query rate drops from 63.5% to 1.9% after direct mutual pairs are deleted. SciDocs, FiQA, and BRIGHT retain more residual nontrivial cyclicity after deletion (10.8%, 30.8%, and 22.0%, respectively), but the residual graphs are structurally much smaller than the full ms1 graphs. Figure 5 shows this before/after decomposition directly.

5.4 Structural Repair Activity

Repair is active almost entirely where cyclicity remains active. Under the primary protocol, ms2 removes zero weight on all four datasets, while ms1 removes nontrivial normalized weight on every dataset and ms1_drop_mutual removes only a small residual amount. Figure 6 reports normalized removed weight directly. As a secondary diagnostic, Table 6 reports qrels-anchored BEW/PIC reductions under the calibrated ms1 graphs. These quantities are computed against the same relevance-derived reference used later for nDCG@ k evaluation, so they should be read as supporting diagnostics rather than independent evidence that structural repair must improve retrieval.

6 DOWNSTREAM QUALITY RESULTS: REPAIR VS. RETRIEVAL

The calibrated structural results above do not answer the retrieval question by themselves. This section therefore evaluates repaired versus unrepaired ranking quality directly under the primary normalized protocol, then asks whether any positive point estimates survive paired testing, multiplicity correction, and influence analysis. We next compare

Table 6. Qrels-anchored structural diagnostics under the primary normalized protocol, ms1 only. ΔBEW and ΔPIC are mean pre-repair minus post-repair changes computed against the same relevance-derived reference ranking used for downstream nDCG evaluation; mean removed weight is the intrinsic repair objective. These diagnostics are therefore descriptive and qrels-anchored rather than independent validation.

Dataset	ΔBEW	ΔPIC	Weight removed
SciDocs	4.250	19.49	8.934
FiQA	3.919	18.24	7.974
HotpotQA	0.563	2.25	1.173
BRIGHT	2.108	9.36	4.370

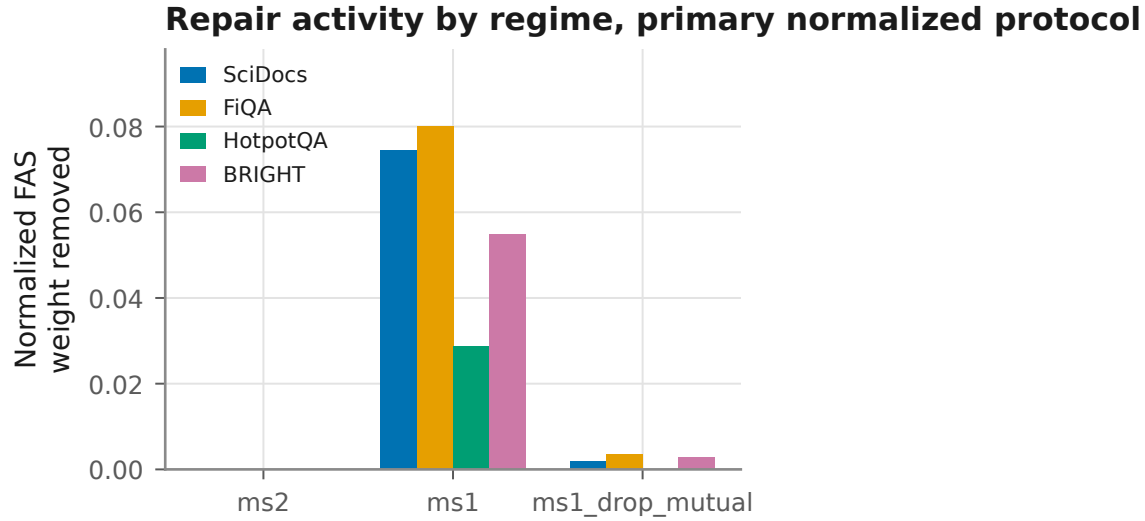


Fig. 6. Mean normalized FAS weight removed (removed weight divided by total graph weight) by vote regime, one bar per dataset, under the primary normalized protocol. ms2 removes exactly 0 on every dataset by construction (it is always acyclic); ms1 removes 0.029–0.080 of total graph weight; ms1_drop_mutual removes only a small residual (≤ 0.004). This is an intrinsic graph diagnostic computed before any retrieval evaluation, not a measure of retrieval benefit.

those calibrated graph-based methods with simple fusion baselines, examine raw-versus-calibrated sign instability, and close with the limited alpha sweep and balance-degeneracy analysis.

6.1 Repaired vs. Unrepaired Retrieval Under the Primary Protocol

Table 7 reports the active ms1 cells for the calibrated Copeland graph and Copeland hybrid. Figure 7 plots the complete calibrated primary grid, including the many zero-effect ms2 and ms1_drop_mutual comparisons. Positive point estimates exist in several cells, but none remains robust after the full testing and sensitivity pipeline reported below.

Most calibrated repaired-versus-unrepaired cells are exactly zero because ms2 is acyclic and ms1_drop_mutual is nearly acyclic on most queries. The substantive activity is concentrated in ms1, but even there the signs are mixed across datasets and methods. SciDocs ms1 is the strongest positive point-estimate case for Copeland, with paired mean deltas of +0.011 for the graph ranking and +0.009 for the hybrid. FiQA ms1 shows a positive Copeland graph estimate but a negative Copeland hybrid estimate. BRIGHT ms1 is near zero for the Copeland hybrid and negative for the

Table 7. Primary normalized repaired-minus-unrepaired $nDCG@k$ for the active ms1 Copeland cells. Intervals are paired percentile bootstrap intervals over common queries; p_{perm} is the paired permutation test p-value on the same query set. No multiplicity adjustment is applied inside this table; corrected results are discussed in the next subsection.

Dataset	Pair	Mean Δ	95% CI	p_{perm}	+/-/=
SciDocs	Copeland graph	+0.011	[0.002, 0.022]	0.020	36/22/62
SciDocs	Copeland hybrid	+0.009	[0.002, 0.017]	0.012	30/10/80
FiQA	Copeland graph	+0.004	[-0.005, 0.011]	0.441	21/10/89
FiQA	Copeland hybrid	-0.005	[-0.013, 0.002]	0.301	16/8/96
HotpotQA	Copeland graph	+0.016	[-0.010, 0.044]	0.314	5/3/44
HotpotQA	Copeland hybrid	+0.012	[0.000, 0.031]	0.253	3/0/49
BRIGHT	Copeland graph	-0.014	[-0.040, 0.006]	0.267	19/13/18
BRIGHT	Copeland hybrid	+0.001	[-0.016, 0.020]	0.910	10/10/30

Repaired – unrepaired $\Delta nDCG$, 95% bootstrap CI (primary normalized protocol)

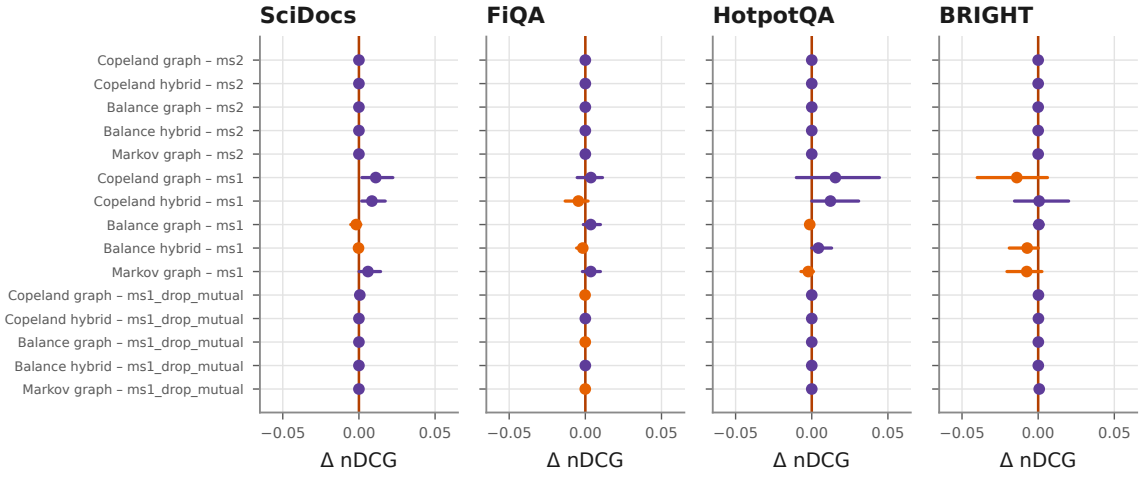


Fig. 7. Repaired-minus-unrepaired $\Delta nDCG$ with paired 95% bootstrap intervals, one panel per dataset and one row per regime $\times$ method pair (15 rows: 5 method pairs $\times$ 3 regimes), with a bold vertical line at zero. Every ms2 and ms1_drop_mutual row collapses to a point exactly at zero; only ms1 rows show visible intervals, and most of those still cross zero. The reader should notice that the handful of intervals not touching zero (e.g., SciDocs and HotpotQA Copeland rows) are exactly the cells discussed in the text below – and that even they do not survive multiplicity correction (Section 6).

Copeland graph. HotpotQA ms1 retains positive Copeland point estimates, but the effect is influence-sensitive and does not survive the robustness checks below. The HotpotQA Markov graph also changes sign relative to the raw-margin ablation, moving from a positive mean in the raw protocol to a negative mean (-0.0023) under the calibrated primary protocol.

6.2 Multiplicity and Influence Robustness

None of the positive repaired-versus-unrepaired cells remains robust after the primary multiplicity correction family is applied. The two smallest raw paired permutation p-values occur on SciDocs ms1 Copeland hybrid ($p = 0.012$) and SciDocs ms1 Copeland graph ($p = 0.020$), but both lose support after Holm and Benjamini–Hochberg adjustment. The

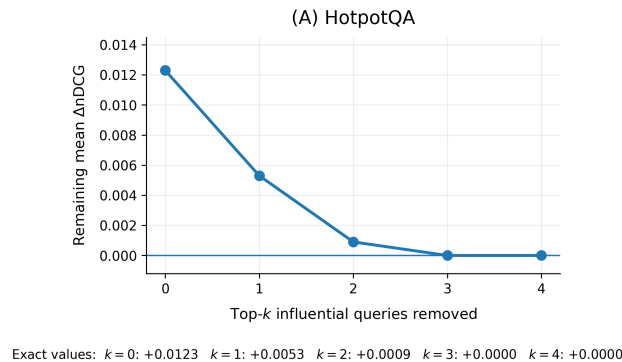


Fig. 8. Remaining mean $\Delta nDCG$ for the calibrated ms1 Copeland-hybrid cell as the top- k most influential HotpotQA queries are removed ($k = 0, \dots, 4$), the more concentrated of the two main positive-point-estimate cases. The mean falls from +0.0123 to exactly 0 after removing the top 3 queries, so the positive estimate is concentrated in a handful of queries. The companion SciDocs cell is less concentrated and is reported numerically in the text below rather than plotted here. Values are exact per-query recomputations, not interpolated.

HotpotQA ms1 Copeland hybrid, which previously supplied the paper’s positive headline, has a fresh paired permutation p-value of approximately 0.253 under the primary normalized protocol and an interval whose lower endpoint is exactly zero rather than strictly positive.

Figure 8 plots this influence pattern directly for the more concentrated HotpotQA cell. HotpotQA’s positive mean is concentrated in a handful of queries. Removing the top one influential query reduces the calibrated ms1 Copeland-hybrid mean delta from +0.0123 to +0.0053; removing the top two reduces it to +0.0009; removing the top three or top four drives the mean to exactly zero. SciDocs is less concentrated: the corresponding Copeland-hybrid mean remains positive after removing the top one, two, three, and four influential queries (+0.0055, +0.0036, +0.0048, and +0.0041), but these nominal positives still do not survive multiplicity correction. The primary evidence therefore supports influence-sensitive positive point estimates in a small number of cells, not a robust positive retrieval effect of repair.

6.3 Raw Versus Calibrated Sign Instability

Calibration changes retrieval conclusions as well as structure. Table 8 lists the five most consequential sign flips between the raw-margin ablation and the primary normalized protocol. These flips are not incidental numeric perturbations: they accompany large shifts in edge retention, edge weighting, and removed-edge overlap, so they show that the paper’s downstream conclusions are not calibration invariant. Figure 9 plots the full raw-versus-calibrated comparison underlying Table 8.

6.4 Comparison with Fixed Aggregation Baselines

The calibrated repaired-versus-unrepaired analysis isolates the marginal effect of repair, but it does not answer whether graph-based methods outperform simple fusion baselines. Table 9 therefore reports dataset macro-averages under the primary ms1 protocol, and Figure 10 shows the corresponding per-dataset baseline comparison. We use dataset-level summaries rather than pooled query $\times$ regime means because the calibrated conclusions are not uniform across datasets.

Simple fusion baselines remain strong after calibration. CombsUM has the highest dataset-macro mean under primary ms1; RRF is also competitive, and both outperform the repaired Copeland graph methods on this summary. The

Table 8. Sign changes between the raw-margin ablation and the primary normalized protocol. The raw protocol uses native heterogeneous scores with fixed thresholds and is shown only as an ablation. The calibrated protocol uses per-query/per-ranker min-max normalization with retention-matched thresholds. Values are repaired-minus-unrepaired mean nDCG@k.

Dataset	Cell	Raw	Calibrated
SciDocs	ms1 Copeland graph	−0.0079	+0.0110
SciDocs	ms1 Copeland hybrid	−0.0018	+0.0085
FiQA	ms1 Copeland graph	−0.0029	+0.0036
BRIGHT	ms1 Copeland hybrid	−0.0045	+0.0006
HotpotQA	ms1 Markov graph	+0.0087	−0.0023

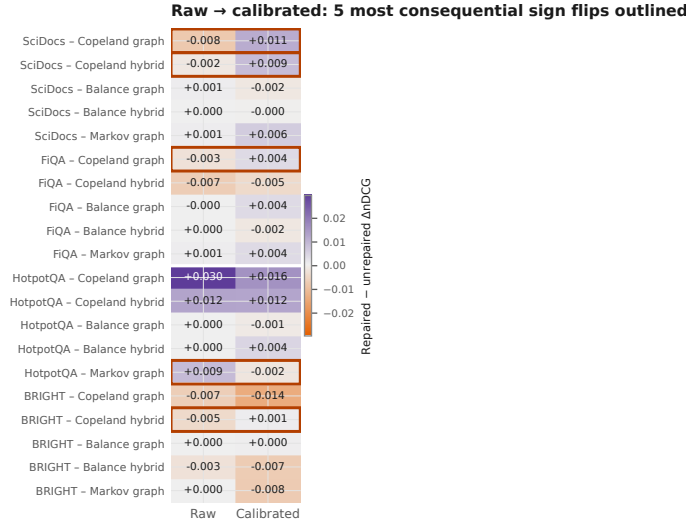


Fig. 9. Repaired-minus-unrepaired Δ nDCG for every dataset $\times$ method pair under ms1, raw-margin ablation versus primary normalized protocol, colored by sign and magnitude (purple positive, orange negative). The five rows outlined are the sign flips Table 8 calls out in the text; most other cells keep the same sign, or flip only at noise-level magnitude on both sides. The reader should take away that a handful of specific, named conclusions reverse under calibration – not that every cell is unstable.

Prior baseline uses the same RRF family on candidate nodes but differs from the standalone RRF baseline in implementation details and exact tie-breaking, so we report both rather than treating them as interchangeable; across the full run, their exact rankings match in only 216 of 6,156 comparable cases. The primary evidence therefore does not support a claim that graph repair makes simple baselines obsolete.

6.5 Alpha Sensitivity and Balance Degeneracy

The small alpha sweep does not rescue a robust positive repair result. Table 10 reports the primary ms1 Copeland-hybrid delta over $\alpha \in \{0.1, 0.3, 0.5, 1.0\}$. SciDocs remains positive for $\alpha \in \{0.3, 0.5, 1.0\}$, but those gains are still not robust after multiplicity correction. HotpotQA stays positive for $\alpha \in \{0.3, 0.5, 1.0\}$, but the effect remains influence-sensitive. FiQA and BRIGHT change sign within this small sweep, showing that stronger graph weight does not induce a stable benefit. Figure 11 plots all four datasets across the full α sweep.

Per-dataset method comparison, ms1, sorted by mean nDCG@k (solid = graph-independent baseline)

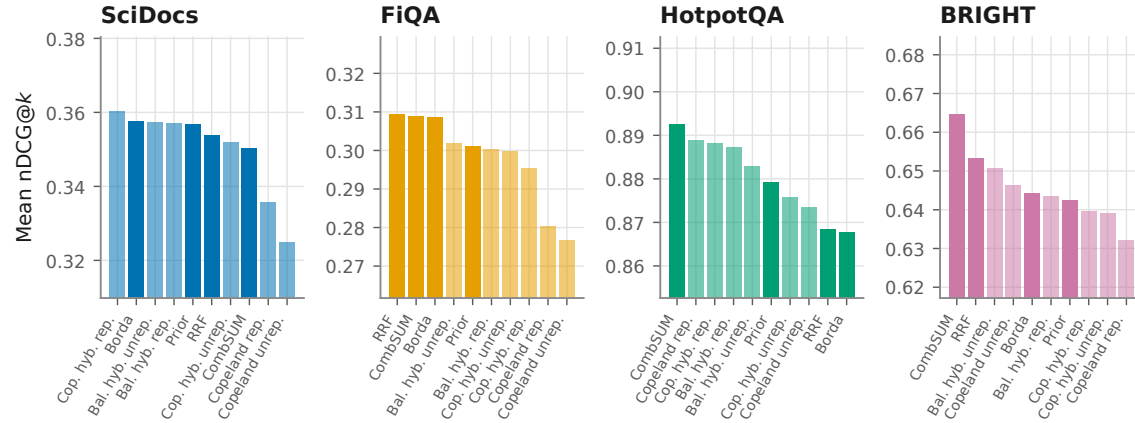


Fig. 10. Mean nDCG@k under ms1, one panel per dataset, methods sorted left to right by value; solid bars are the four graph-independent baselines (Prior, RRF, CombSUM, Borda), translucent bars are graph-dependent methods. Note the per-dataset y -axis: each panel is scaled to its own score range so within-dataset ordering is visible, so only within-panel comparisons are meaningful, not bar heights across panels. A graph-independent baseline is within the top three methods on every dataset (CombSUM leads on HotpotQA and BRIGHT, RRF on FiQA), while on SciDocs a repaired Copeland-hybrid edges out CombSUM by a narrow margin – consistent with Table 9’s macro-average ranking, which pools across all four datasets rather than reporting any single one.

Table 9. Dataset-macro comparison under the primary normalized protocol, ms1 only. Mean nDCG is averaged across the four datasets; average rank is the mean within-dataset rank by mean nDCG. Because Prior and RRF use closely related but not identical implementations, both are reported separately rather than treated as the same baseline.

Method	Macro mean nDCG	Avg. rank
CombSUM	0.554	3.25
Balance hybrid unrepared	0.548	4.75
Balance hybrid repaired	0.547	6.00
RRF	0.546	5.50
Copeland hybrid repaired	0.546	6.50
Prior	0.545	6.75
Borda fusion	0.545	6.00
Copeland hybrid unrepared	0.542	9.00
Balance repaired	0.539	8.25
Balance unrepared	0.539	8.25
Markov unrepared	0.537	9.50
Markov repaired	0.537	9.25
Copeland repaired	0.534	11.00
Copeland unrepared	0.530	11.00

The balance degeneracy analysis explains why many balance results remain exactly or nearly zero despite nontrivial graph changes. Under the primary ms1 protocol, the graph changes on 119/120 SciDocs queries, 118/120 FiQA queries, 33/52 HotpotQA queries, and 46/50 BRIGHT queries. Those changes do not consistently propagate to the evaluation

Table 10. Copeland-hybrid alpha sensitivity under the primary normalized protocol, ms1 only. Entries are repaired-minus-unrepaired mean nDCG@k; this is a small sensitivity sweep, not a tuned hyperparameter search.

Dataset	$\alpha = 0.1$	$\alpha = 0.3$	$\alpha = 0.5$	$\alpha = 1.0$
SciDocs	-0.0022	+0.0085	+0.0097	+0.0083
FiQA	+0.0006	-0.0046	+0.0010	+0.0038
HotpotQA	-0.0008	+0.0123	+0.0120	+0.0101
BRIGHT	-0.0013	+0.0006	-0.0074	-0.0071

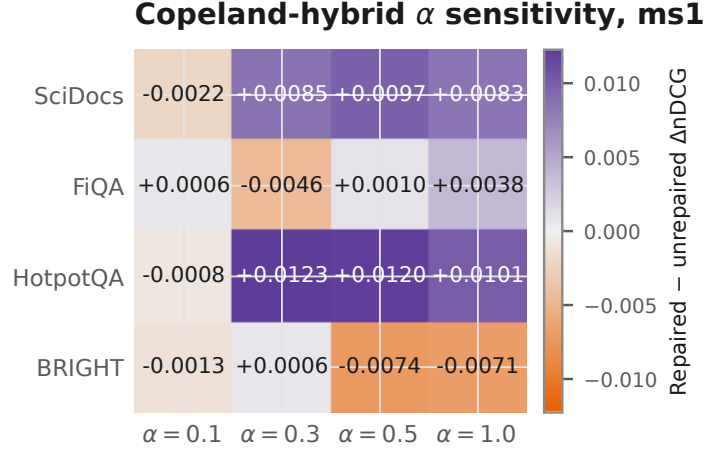


Fig. 11. Copeland-hybrid repaired-minus-unrepaired mean Δ nDCG under ms1 as the graph-component weight α varies over $\{0.1, 0.3, 0.5, 1.0\}$, colored by sign and magnitude. SciDocs and HotpotQA stay positive for $\alpha \geq 0.3$; FiQA and BRIGHT change sign within this small sweep. The reader should take away that increasing the graph component's weight does not produce a consistent direction of effect across datasets; this is a small sensitivity sweep, not a tuned hyperparameter search or a multiplicity-adjusted test.

metric: the full balance-hybrid ranking changes on only 68, 68, 7, and 20 queries respectively, and nDCG changes on only 8, 5, 1, and 8 queries. In other words, the graph can change, the balance scores can change, and even the top- k order can change without altering the relevance sequence scored by nDCG@ k . The balance results are therefore not mysterious nulls; they are a direct consequence of downstream propagation failure from graph repair to the evaluated prefix.

7 SECONDARY ANALYSES AND SCOPE CHECKS

The calibrated core evidence of this paper is contained in Sections 5 and 6. The four topics below are retained only as scoped secondary notes because they do not currently meet the same evidentiary standard as the calibrated mechanical evaluation.

7.1 Rule-Based Outcome Taxonomy

A separate raw-margin failure-mining corpus of 1,020 records was previously labeled with a six-way rule-based outcome taxonomy. That taxonomy is not manually verified: it is an automated rule cascade whose final labels depend heavily on branch precedence, and its counts were never regenerated under the primary normalized protocol. Because normalization and vote construction change retained edges, removed edges, and repaired-versus-unrepaired signs, those raw-margin counts are not appropriate as evidence for the present study.

We therefore do not treat that taxonomy as part of the main empirical narrative. A *rule-based outcome taxonomy* may still be useful as future descriptive analysis, but it would need to be regenerated under the primary protocol before interpretation. Such a taxonomy would remain a retrospective outcome classification, not a validated predictive mechanism available before repair.

7.2 Protocol Audit of Stored LLM Judgments

We also inspected stored real-LLM judgment corpora to determine whether they provide an independent validation of the calibrated mechanical study. They do not currently support that role. The stored corpora exhibit parser defaults that silently map ambiguous or malformed outputs to a fixed label, measurable provider-dependent position bias, incomplete raw-text and exclusion logging, and protocol differences across providers and method families. These issues are serious enough that apparent cyclicity in a naively constructed LLM-derived graph can be dominated by pipeline artifacts rather than by genuine model disagreement.

Quantitatively, a stored pairwise-judgment corpus collected from the Cohere and Azure OpenAI APIs on FiQA, HotpotQA, and BRIGHT shows the following. The shared response parser silently defaults ambiguous or malformed outputs to a fixed label on 1.1% of 12,020 responses with preserved raw text (0.0% on FiQA for both providers; up to 3.8% on HotpotQA for Cohere). Among well-formed, unambiguous responses, both providers show statistically significant position bias, but in *opposite* directions: Azure prefers the first-shown candidate 53–58% of the time across the three datasets, while Cohere prefers the second-shown candidate 61–70% of the time (exact binomial $p < 0.01$ in every provider–dataset cell); pooling the two providers would largely cancel this out and hide both effects. Forward and reverse presentations of the same candidate pair agree with each other only 59–85% of the time depending on provider and dataset (Cohen’s κ 0.27–0.71, fair to substantial). In a preference graph built directly from these judgments (not the mechanical graph used elsewhere in this paper), removing edges produced by the parser’s default-label rule and by the position-bias-driven disagreement default reduces measured cyclicity by 91–100% (from a 31–62% cyclic range down to 0–5%), indicating that most of that graph’s apparent cyclicity reflects these pipeline defaults rather than genuine model intransitivity. None of these figures alter the mechanical-graph statistics reported in Sections 5 and 6, which do not depend on any LLM judgment; they bound what a future LLM-judgment-derived graph-construction study would need to resolve before its cyclicity could be interpreted at face value.

For that reason, we do not pool the stored LLM corpora with the primary mechanical-vote evaluation or treat them as a parallel replication. The main lesson from these records is methodological: pairwise-judgment pipelines

can manufacture apparent inconsistency if prompt symmetry, abstention behavior, parser handling, and exclusion accounting are not controlled carefully. Accordingly, we retain this material only as a bounded protocol-quality check.

A future LLM-based study intended to support the paper’s central claims would need randomized candidate order, symmetric A/B and B/A prompting, abstention on malformed or ambiguous outputs, no default-label fallback, complete exclusion logs, and matched query/method protocols across providers.

7.3 Implementation Cost Note

We retain only a bounded implementation-cost observation. The calibrated core pipeline is CPU-only and practical on the graph sizes studied here, but the available runtime and memory evidence comes from a single unbenchmarked local machine with coarse resident-set-size snapshots and some timings near the instrumentation floor. We therefore do not present efficiency as a substantive contribution or make cross-system scaling claims. Repair cost depends on graph size, density, and SCC structure, and a controlled hardware benchmark would be required before stronger conclusions could be justified.

7.4 Planned CARB Resource

CARB is a planned feature-only companion resource, not a released benchmark in this review draft. No empirical claim depends on it. Any future release would need a review-ready package, corrected scope statistics, and licensing review consistent with the source datasets’ redistribution terms.

8 DISCUSSION

Sections 5–7 support a conclusion more specific than a naive “repair either helps or it does not” framing: preference-graph repair cannot be interpreted independently of the score-normalization and vote-construction protocol that creates the graph. Under the primary normalized retention-matched protocol, repair still changes graph structure, but its downstream retrieval effect remains construction-sensitive, method-dependent, and not robustly positive across the tested benchmarks.

Normalization is part of the model, not preprocessing trivia. The preference graph inherits whatever score scale is used to construct pairwise margins. Under the unnormalized raw-margin construction, those margins were not comparable across BM25, TF-IDF, and MiniLM, and BM25 contributed almost all conditional edge weight when it participated. The primary normalized protocol substantially reduced that scale domination and, with it, changed which votes survived, which edges were retained, and which edges were removed during repair. The practical lesson is methodological: conclusions about graph repair are inseparable from the normalization and threshold semantics used upstream. We do not claim that per-query, per-ranker min-max normalization is universally optimal, only that it is the best-supported primary protocol in this study because it materially reduced score-scale distortion and enabled a controlled comparison against the raw ablation.

Structural success does not guarantee retrieval gain. FAS repair (Eq. (9)) optimizes a graph-internal objective, whereas nDCG evaluates ranked retrieval against qrels. That distinction matters empirically: the calibrated experiments still show nonzero repair activity in the cyclic regimes, yet structural improvement does not imply a better final ranking. Depending on the query, extraction rule, and fusion stage, a repaired graph can be retrieval-inactive, evaluation-invisible, neutral, or harmful. The qrels-anchored diagnostics in Section 5 are useful for interpretation, but they should not be treated as independent validation of the graph objective itself. Structural consistency and retrieval effectiveness remain related but distinct empirical targets.

Mutual contradictions and longer-cycle structure should be reported separately. The calibrated results also change how cyclicity should be read. Apparent graph cyclicity is substantial under the primary protocol before mutual pair deletion, but much of it arises from direct bidirectional contradictions rather than from longer cyclic structure. After deleting mutual pairs, the mean cyclic-query rate drops markedly, leaving a smaller but still nonzero residual of nontrivial cyclicity. This does not make the remaining cycles unimportant, but it does mean that total ms1 cyclicity should not be interpreted automatically as genuine Condorcet-style nontransitivity. Future work should report both direct mutual contradiction and post-deletion cyclicity by default.

Repair effects are method- and calibration-dependent. The same repaired graph does not propagate uniformly through Copeland, Markov, balance, and the hybrid variants. Some calibrated cells change sign relative to the raw package, and the balance-family zero-delta pattern in this study reflects propagation failure through the ranking pipeline in this study, not a general impossibility result for balance-based repair. The hybrid results similarly show that fusion can either dampen noisy graph changes or suppress potentially meaningful ones, depending on how much signal survives calibration and how strongly the prior ranking dominates the final score. Positive point estimates therefore require more than a favorable mean: in this study, none remains robust once paired testing, influence analysis, and multiplicity correction are applied together. HotpotQA is the clearest example of a positive-looking but non-robust cell: its calibrated ms1 Copeland-hybrid mean is positive, yet the effect is concentrated in a few influential queries, the paired permutation test is not compelling, and the cell does not survive Holm or Benjamini–Hochberg adjustment.

Simple baselines remain important under calibrated graphs. The present study also does not support graph repair as a superior default retrieval strategy. Reciprocal-rank fusion and CombSUM remain strong comparators under the calibrated protocol, and the prior ranking continues to absorb much of the useful signal in the hybrids. At the same time, these comparisons should be read with the candidate-pool design in mind: the experiments are centered on an RRF-family pooling scheme, and Prior and the RRF baseline, although related, are not interchangeable rows. The right takeaway is therefore cautious rather than triumphalist: simple fusion baselines remain strong under the calibrated protocol, and graph-dependent methods must justify their added complexity on a paired, dataset-specific basis.

Implications for future preference-graph studies. Preference-graph experiments should report calibration, threshold interpretation, common-query pairing, mutual-pair prevalence, residual nontrivial cyclicity, repair objective, ranking-extraction method, influence diagnostics, and multiplicity correction as first-class design choices rather than hidden preprocessing details. Raw-score constructions can still be informative as ablations, but not as primary evidence when the underlying rankers operate on heterogeneous score scales. This paper therefore does not offer a validated predictive selector for when repair will help. It offers a narrower but more defensible conclusion: repair outcomes depend materially on the graph-construction process, and any future claim about repair effectiveness should make that dependence explicit.

9 LIMITATIONS

This study’s limitations fall into several categories, detailed below.

Normalization remains a design choice, not a proven optimum. The primary protocol uses per-query, per-ranker min–max normalization with retention-matched thresholds because that combination substantially reduced score-scale domination and supported the most defensible controlled comparison in this study. It is not the only plausible normalization. Other transformations in the pilot changed graph structure and some downstream signs, so the present paper demonstrates construction sensitivity rather than proving that one normalization is universally best.

Candidate pooling and baseline roles are RRF-centered. The experiments use an RRF-family candidate-pool construction and then compare graph methods against Prior, RRF, CombSUM, Borda, and related variants within that setting. Because candidate selection, Prior, and the RRF baseline occupy different implementation roles, conclusions may differ under independently constructed pools or different prior rankings. The paper documents these distinctions, but it does not establish pool-invariant behavior.

Nontrivial cyclicity is smaller than headline cyclicity. Under the primary protocol, cyclicity after removing direct mutual pairs is materially lower than cyclicity before deletion. That means some regimes offer limited opportunity for repair to act on longer-cycle structure, and the paper does not establish how the same methods would behave on corpora with naturally high nontrivial cyclicity.

Retention-target sensitivity changes structure more than retrieval robustness. The retention-matching procedure (Section 3.3) is an experimental control for comparing raw and normalized graph construction at approximately matched retention, not a universally optimal threshold-selection method. A sensitivity sweep of alternative retention/threshold policies shows that graph density, cyclicity, and removed-edge patterns change materially across policies, and some repaired-versus-unrepaired cell signs flip. Across those policies, however, no positive repaired-versus-unrepaired nDCG cell survives Holm correction. We therefore retain raw-reference retention matching as the primary protocol and interpret structural diagnostics as policy-sensitive, while the main multiplicity-corrected retrieval conclusion remains robust under the tested alternatives.

The repair algorithm is heuristic and order-sensitive. Our main repair procedure is greedy cycle peeling with no approximation guarantee, and this is the only repair used to produce the primary evidence in Sections 5–6. As a robustness check (Section 4.4), we additionally solved the exact minimum-weight feedback arc set problem with an open-source ILP solver on the complete 1,025-query primary analysis: the exact solution differs from greedy’s on most cyclic queries and removes materially less weight, but this does not change which repaired-versus-unrepaired retrieval cells survive multiplicity correction. In principle, cycle-discovery order, insertion order, and alternative optimization strategies could still produce different greedy-repaired graphs and therefore different downstream rankings; the exact-solver check bounds, but does not eliminate, this concern for the primary greedy results.

Statistical power and influence remain limiting factors. Several cells have modest query counts, and some positive-looking means are concentrated in a small number of influential queries. The paper addresses this with paired permutation tests, bootstrap intervals, leave-one-out analysis, influence removal, and multiplicity correction, but those diagnostics also reinforce the main caution: no positive repaired-versus-unrepaired effect remains robust after Holm or Benjamini–Hochberg correction.

Metric scope is narrower than graph scope. nDCG is the main retrieval metric, and its cutoff determines whether some ranking changes are visible at all. MAP and MRR are reported only as secondary robustness checks and are not part of the primary multiplicity-corrected nDCG family; pool-truncated Recall@ k is not used as a confirmatory metric. The qrels themselves are incomplete and dataset-dependent, and BEW/PIC are computed from the same relevance evidence rather than from an independent source of graph truth. The paper therefore does not claim that improved structural consistency implies a general improvement in user utility beyond the measured retrieval setting.

The stored LLM records are limited as evidence. As Section 7.2 now makes explicit, parser defaults, position bias, incomplete logging, and protocol differences prevent the current LLM artifacts from supporting a clean claim about genuine model nontransitivity or from serving as independent validation of the calibrated mechanical study.

External validity remains bounded. The primary evidence is drawn from four retrieval benchmarks, three mechanical rankers, and a specific set of vote regimes, extraction methods, and fusion choices. We do not claim that the

same calibrated conclusions will hold unchanged for other domains, ranker families, pairwise-preference sources, or evaluation metrics.

External companion resources remain optional. CARB is not released; evaluate this manuscript on the primary evidence reported here (Section 11), not on a promised companion release.

10 CONCLUSION

This study examines not only whether preference-graph repair affects retrieval, but also whether that conclusion is stable under defensible choices for normalizing and aggregating heterogeneous ranking scores. The central finding is methodological. Raw heterogeneous ranker margins severely distorted weighted graph construction, producing near-total BM25 dominance whenever BM25 participated. Replacing those raw margins with per-query, per-ranker min-max normalization and retention-matched thresholds materially changed retained edges, cyclicity, removed edges, and several repaired-versus-unrepaired signs.

Under this primary normalized protocol, repair still improves internal graph consistency in the active regimes, but structural consistency and retrieval effectiveness do not move together reliably. Positive repaired-versus-unrepaired point estimates remain in some cells, yet none survives the combined scrutiny of paired testing, influence analysis, and multiplicity correction. The paper therefore does not support a construction-invariant claim that repair helps, nor does it support a single robust positive exception. It instead shows that repair outcomes are sensitive to normalization and vote construction, and that simple fusion baselines remain strong comparators under the normalized protocol.

The practical implication is modest but important: preference-graph repair should be evaluated together with the score normalization, threshold semantics, mutual-pair handling, and ranking-extraction choices that determine the graph on which repair operates. Future work may yet find settings where repair yields a robust retrieval gain, but this study shows that such a claim cannot be treated as independent of graph construction.

11 DATA AVAILABILITY AND REPRODUCIBILITY

Data availability. SciDocs, FiQA, HotpotQA, and BRIGHT are publicly available from their original sources [4, 15, 19, 23]. This paper redistributes no raw third-party document collection. CARB remains planned only (Section 7.4).

Artifact availability. The pipeline in Sections 3–4 is packaged as a reproducible code-and-data artifact. This double-anonymous draft does not include an author-identifying public repository URL, and no verified anonymized mirror URL is claimed here. A scrubbed anonymous artifact should be attached through the venue submission system before review (not by email request). The package contains stored ranker scores, query-ID manifests, qrels/score hashes, primary normalized and raw-ablation outputs, tables/figures, and commit manifests. The primary evidence set is the normalized retention-matched package; the raw-margin package is an ablation only.

Reproducibility. Mechanical results in Sections 5–6 regenerate from stored intermediates with the seeds, solver settings, and eligibility rule in Section 4. Upstream retrieval and paid APIs need not be rerun. Stored LLM records support protocol-quality checks only. The review artifact includes a dependency list (`requirements.txt`) together with the code snapshot used for mechanical regeneration.

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